

Experiment 3

AIM: Develop Expense Tracking Module with Exception Handling.

Source: Practical 3.cs

Code:

```
using System;

using System.Collections.Generic;

using System.Linq;

using System.Text;

using System.Threading.Tasks;

namespace prac3

{

    class Expense

    {

        public int expenseId;

        public string expenseName;

        public string category;

        public string expenseDate;

        public double amount;

        public Expense()

        {

            Console.WriteLine("Expense Object Created");

        }

        public void DisplayExpense()

        {
```

```
Console.WriteLine("\nExpense Details");

Console.WriteLine("Expense ID   : " + expenseId);

Console.WriteLine("Expense Name : " + expenseName);

Console.WriteLine("Category    : " + category);

Console.WriteLine("Expense Date : " + expenseDate);

Console.WriteLine("Expense Amount : " + amount);

}

}

class Program
{
    static void Main(string[] args)
    {
        Expense exp = new Expense();

        try{

            Console.Write("Enter Expense ID: ");

            exp.expenseId = int.Parse(Console.ReadLine());

            if (exp.expenseId <= 0)
            {
                throw new Exception("Expense ID must be greater than 0.");
            }

            Console.Write("Enter Expense Name: ");

            exp.expenseName = Console.ReadLine();

            if (string.IsNullOrEmpty(exp.expenseName))

                {
```

```
        throw new Exception("Expense Name cannot be empty.");
    }

    Console.WriteLine("Enter Category: ");

    exp.category = Console.ReadLine();

    if (string.IsNullOrEmpty(exp.category))
    {
        throw new Exception("Category cannot be empty.");
    }

    Console.WriteLine("Enter Expense Date (dd/mm/yyyy): ");

    exp.expenseDate = Console.ReadLine();

    if (string.IsNullOrEmpty(exp.expenseDate))
    {
        throw new Exception("Expense Date cannot be empty.");
    }

    Console.WriteLine("Enter Expense Amount: ");

    exp.amount = double.Parse(Console.ReadLine());

    if (exp.amount <= 0)
    {
        throw new Exception("Expense Amount must be greater than 0.");
    }

    exp.DisplayExpense();
}

catch (FormatException)
{

```

```
        Console.WriteLine("Invalid Input! Please enter numeric values where required.");
    }

    catch (Exception ex)

    {

        Console.WriteLine("Error: " + ex.Message);

    }

    finally

    {

        Console.WriteLine("\nExpense Tracking Completed.");

    }

    Console.ReadKey();

    }
}
```

Output:

```
Expense Object Created
Enter Expense ID: 5
Enter Expense Name: food
Enter Category: food
Enter Expense Date (dd/mm/yyyy): 27/7/26
Enter Expense Amount: 2000

Expense Details
Expense ID      : 5
Expense Name    : food
Category        : food
Expense Date    : 27/7/26
Expense Amount  : 2000

Expense Tracking Completed.
```

GitHub URL: <https://github.com/shrey2250/.net-practicals.git>